

$$y = g(x) \quad x = g^{-1}(y) = h(y)$$

$$g(g^{-1}(x)) = x$$

$$g'(g^{-1}(x)) (g^{-1})'(x) = 1$$

$$g'(y) = \frac{1}{(g^{-1})'(x)}$$

$$g(h(\vec{x})) = \vec{x}$$

$$Dg(h(\vec{x})) Dh(\vec{x}) = I_{d \times d}$$

$$Dg(y) = [Dh(\vec{x})]^{-1}$$

$$X = AZ + \mu$$

$\xrightarrow{d \times 1} \quad \xrightarrow{d \times d} \quad \xrightarrow{d \times 1}$

$$X_i = A^i Z + \mu_i$$

$\xrightarrow{i^{\text{th}} \text{ row of } Z}$

$$= \sum_{j=1}^d A_{ij} Z_j + \mu_i = A_{ik} Z_k + \mu_i + \sum_{\substack{j=1 \\ j \neq k}}^d A_{ij} Z_j$$

$$\frac{\partial X_i}{\partial Z_k} = A_{ik}$$

$$\frac{\partial X}{\partial Z} = \begin{bmatrix} \frac{\partial X_1}{\partial Z_1} & \frac{\partial X_1}{\partial Z_2} & \dots & \frac{\partial X_1}{\partial Z_d} \\ \frac{\partial X_2}{\partial Z_1} & & & \\ \vdots & & & \\ \frac{\partial X_d}{\partial Z_1} & & & \frac{\partial X_d}{\partial Z_d} \end{bmatrix}$$

$A_{11} \quad A_{12} \quad A_{1d}$
 $A_{21} \quad A_{d1} \quad A_{dd}$

$$f_Z(z) = \frac{e^{-\frac{1}{2} z^T z}}{(2\pi)^{d/2}}$$

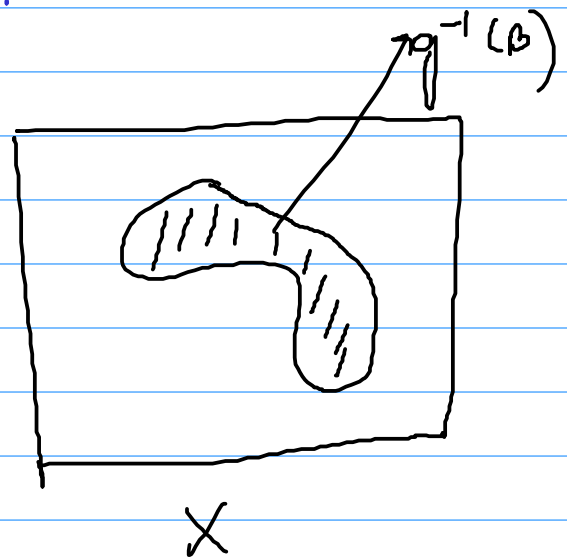
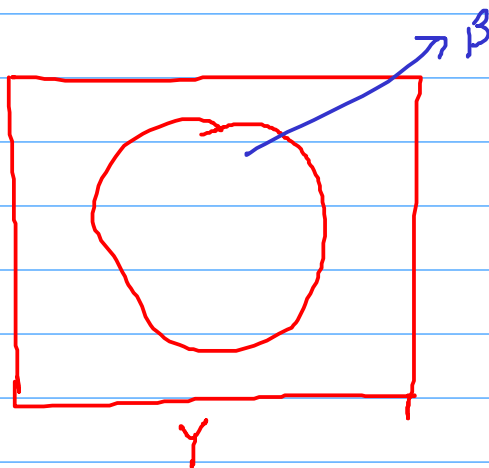
$$f_X(x) = \frac{f_Z(\bar{A}'(x-\mu))}{|A|}$$

$$= \frac{\exp\left(-\frac{1}{2} (\bar{A}'(x-\mu))^T \bar{A}'(x-\mu)\right)}{(2\pi)^{d/2} |A|}$$

$$= \frac{\exp\left(-\frac{1}{2} (x-\mu)^T (\bar{A}')^T \bar{A}' (x-\mu)\right)}{(2\pi)^{d/2} |A| |C|^{1/2}}$$

$$C = AA^T \quad C^{-1} = (AA^T)^{-1} = (\bar{A}')^{-1} \bar{A}'$$

$$|C| = |A| |A^T| = |A|^2$$



$$P(Y \in B) = \int_B f_Y(y) dy$$